Script-Controlled ACL: Restrict Record Access Based on Field Value

Project Design Phase: Problem – Solution Fit Template

Date: October 3, 2026	**Maximum Marks:** 4 Marks
Team ID: SWTID-2026-8089	**Project Name:** Script-Controlled ACL

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer's problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why.

Purpose:

- **Protect student records:** Use script-controlled ACLs in ServiceNow so that users see, create, edit and delete only the records that their role and Branch value allow, while administrators keep full access.
- **Solve complex problems in a way that fits the state of your customers:** Remove the risk of open table access by combining the roles bb1 to bb4, the data condition Branch is EEE and an ACL script, with simple verification through impersonation.
- **Succeed faster and increase your solution adoption by tapping into existing mediums and channels of behavior:** Use the standard ServiceNow web UI that users already work in (lists, forms, New button, inline edit), so no new tool or training is needed.
- **Sharpen your communication and marketing strategy with the right triggers and messaging:** Apply the rules at the moment a user opens a list or record, and give a clear outcome such as "No records to display" for users without the role.
- **Increase touch-points with your company by finding the right problem-behavior fit and building trust by solving frequent annoyances, or urgent or costly problems:** Remove the worry about student data exposure and about accidental edits or deletions by users who should not have that permission.
- **Understand the existing situation in order to improve it for your target group:** Replace open, unrestricted table access and manual checking with automatic, role-based and field-based record security.

TEMPLATE:

Problem-Solution fit canvas 2.0

PURPOSE / VISION: Protect student records in ServiceNow with script-controlled ACLs so users see and change only the records their role and branch allow, while administrators keep full access.

1. CUSTOMER SEGMENT(S) CS

Primary: ServiceNow administrators / security administrators responsible for protecting student records.

Secondary: EEE branch users and faculty with the roles bb1 to bb4 who view and manage records, and other users who have no role.

6. CUSTOMER CONSTRAINTS CC

ACLs can be created only after elevating to the security_admin role.

Rules must work with the roles, users and data already in the instance.

Administrators must keep full access to all records.

Each rule must be tested by impersonating every user type.

5. AVAILABLE SOLUTIONS AS

Default table access: any user with basic access can view records; no protection.

Role-only ACLs: block users without a role but cannot filter by field value such as Branch.

Client-side hiding (UI policies): easy to bypass and not real security.

Manual checking or separate lists per branch: slow, error-prone and hard to maintain.

2. JOBS-TO-BE-DONE / PROBLEMS J&P

Job-to-be-done: Control which records each user can read, create, edit and delete based on role and Branch value.

Problems:

Any user with basic access can view, edit, create or delete sensitive student records.

No record-level rule checks the Branch field value.

Read, create, write and delete permissions are not separated by role.

9. PROBLEM ROOT CAUSE RC

The table has no record-level rules that check a user's role together with a field value (Branch).

Read, create, write and delete operations are not tied to separate roles (bb1 to bb4).

Without a script-controlled ACL, access is not denied by default.

7. BEHAVIOUR BE

Users open the Institution Details list directly (u_institution_details.list) and expect to see and manage their records.

Administrators verify access by impersonating the EEE user, a user without roles and the admin.

When a permission is missing, users notice a missing New button, no inline edit, no Delete action, or an empty list.

3. TRIGGERS TR

A user opens the Institution Details list or a record form.

A new user, role or branch is added to the system.

A data-protection requirement or audit review of student data.

10. YOUR SOLUTION SL

Script-controlled ACLs on u_institution_details:

READ: role bb1, data condition Branch is EEE, advanced script that allows admin and bb1 users.

CREATE: role bb2 (New button).

WRITE: role bb3 (edit records).

DELETE: role bb4 (delete records).

The rules are evaluated automatically on every access and verified by impersonating the EEE user, a user without roles and the admin.

8. CHANNELS of BEHAVIOUR CH

8.1 ONLINE

ServiceNow web UI: Institution Details list and forms, Access Control (ACL) list, Impersonate User option.

4. EMOTIONS: BEFORE / AFTER EM

Before: Worried and concerned about data exposure and unauthorized changes.

After: Confident and relieved because bb1 users see only EEE records, users without roles see none, and the admin sees all.

8.2 OFFLINE

Discussions with faculty and branch coordinators about who should access which records; institution data-protection policy.

References:

1. https://www.ideahackers.network/problem-solution-fit-canvas/
2. https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe